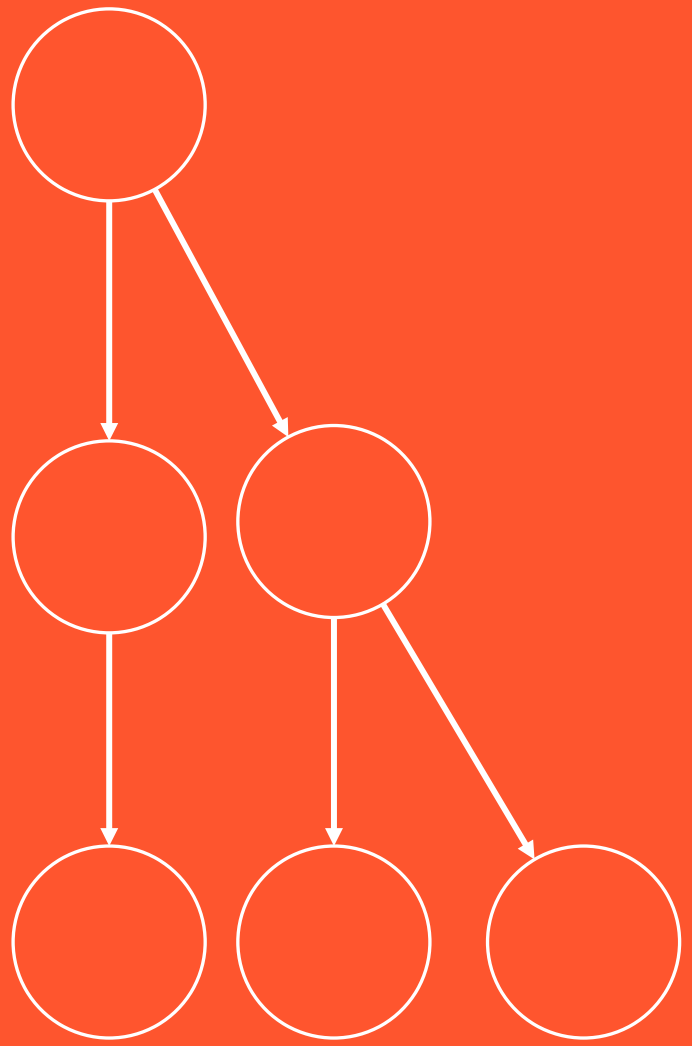


BCOG 100

Planning and Decision-making
Lecture 12.3 — Brains, Genes, and Intelligence

UNIVERSITY OF ILLINOIS URBANA-CHAMPAIGN

To Start: Looking Inside the Brain

In your group, three minutes.

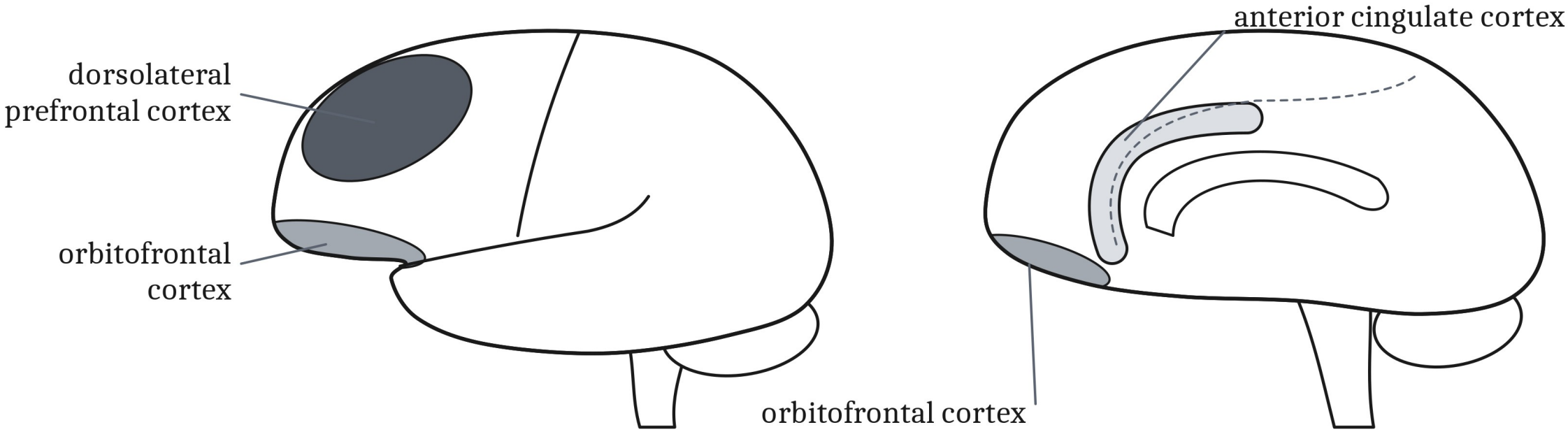
In the second lecture of this module, we described three accounts of why scores on different tests go together.

What were they, and how do they differ? If you could look inside the brain, what would you look for to tell them apart?

The Prefrontal Cortex and Its Parts

Prefrontal cortex: the cortex at the front of the frontal lobe, ahead of the areas that control movement.
Dorsolateral prefrontal cortex: the upper part of the prefrontal cortex on the outer side of each hemisphere. Damage there can leave a person unable to change the rule they follow, without lowering their IQ more than damage elsewhere.

Three parts of the prefrontal cortex



outer side, left hemisphere, front at left

inner surface, right hemisphere, front at left

schematic; boundaries approximate

Control as a Bias

One proposal: the prefrontal cortex maintains the goal and the rules of a task, and uses them to bias processing elsewhere. On this account, the practiced response loses a competition in other regions, and once a task is well practiced, the prefrontal cortex is no longer needed.

Items like those in Stroop's 1935 experiments

						for 100 items
name the ink color	**RED**	**GREEN**	**BROWN**	**BLUE**	**PURPLE**	**about 110 s**
name the color						**about 63 s**

design and times from Stroop (1935), pp. 647–650; averages of 110.3 and 63.3 seconds, Table III

Frontal Patients Who Choose Badly

After frontal damage, a patient known as EVR scored in the top few percent on an IQ test, and still decided badly in daily life. A test score can miss abilities that such patients lose.

On a test, a person...	Decisions in life require...
solves short, well-defined problems, one at a time	organizing several goals over time
is told by the examiner what counts as success	setting one’s own goals
starts when the examiner says to start	bringing choices to an end

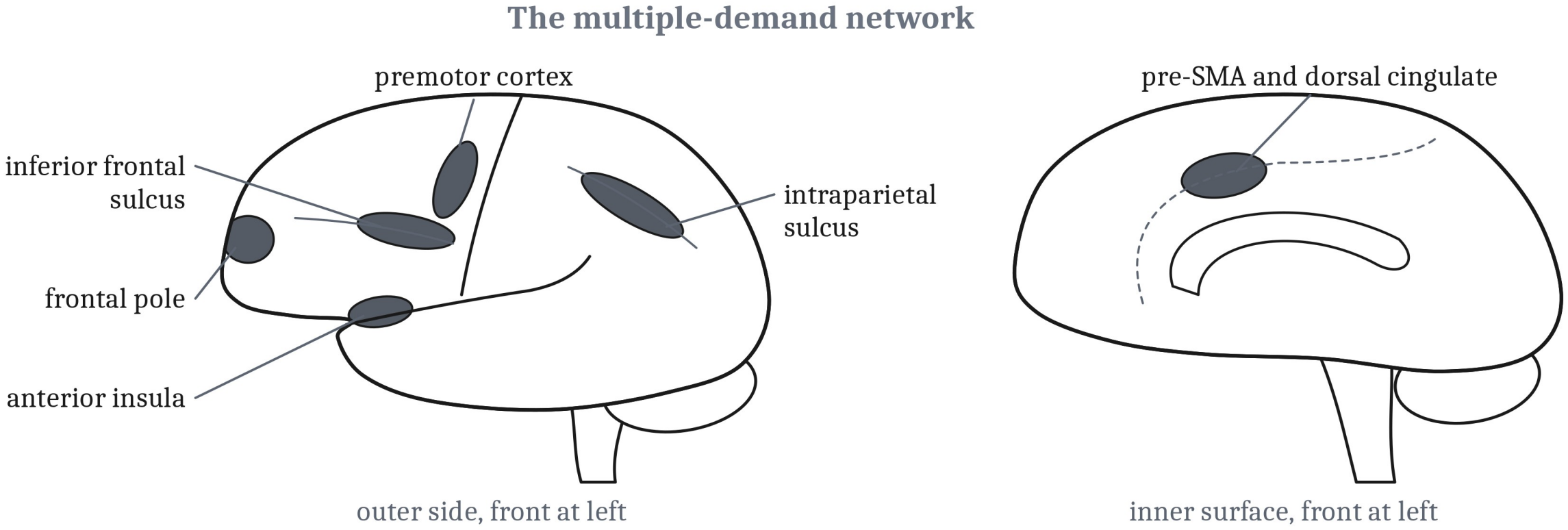
Value in the Brain

Many neurons in the **orbitofrontal cortex**, the underside of the prefrontal cortex just above the eyes, fire in proportion to the value of an option, as measured from the animal's own choices. **Value**: an estimate of how well things will go from a given situation onward.

Step of a decision based on value	What it corresponds to
Representing the options	Not discussed in this module
Assigning each a value	Orbitofrontal value cells
Selecting an action	May correspond to action selection in the basal ganglia (Module 6); little is known
Evaluating the outcome	Not discussed in this module
Learning	Reward prediction errors (Module 5)

Intelligence in the Brain

The **multiple-demand network**: a set of regions in the frontal and parietal lobes of both hemispheres that become more active as many different kinds of task become harder. Across many studies, brain volume correlates with IQ at about 0.24, so it accounts for about 6% of the differences among people.



the same regions respond more to the harder version of many different tasks

schematic, after Duncan (2010), Fig. 1, and Fedorenko et al. (2013); locations approximate

What Heritability Is, and Is Not

Heritability: the proportion of the variation in a trait, in a particular population and set of environments, that is associated with genetic variation among its members.

Heritability is...	**Heritability is not...**
a statistic about a population	a statement about a person
the share of variation associated with genetic variation	by itself, evidence of how, or even whether, genes act on a trait
defined for a particular population and set of environments	a fixed property of a trait

Twin Studies

A **twin study** compares how similar identical twins are with how similar fraternal twins are, on the assumption that the two kinds of twins share the environments that matter for the trait to the same degree. The assumption is disputed, so the twin estimates may be somewhat too high.

What was estimated	Estimate
Correlation, identical twins (higher cognitive functions)	about 0.71
Correlation, fraternal twins (higher cognitive functions)	about 0.44
Heritability: twice the difference	about 0.54
Heritability of IQ, one sample of young U.S. twins, poor families	close to zero
Heritability of IQ, one sample of young U.S. twins, affluent families	high

Genes Measured Directly

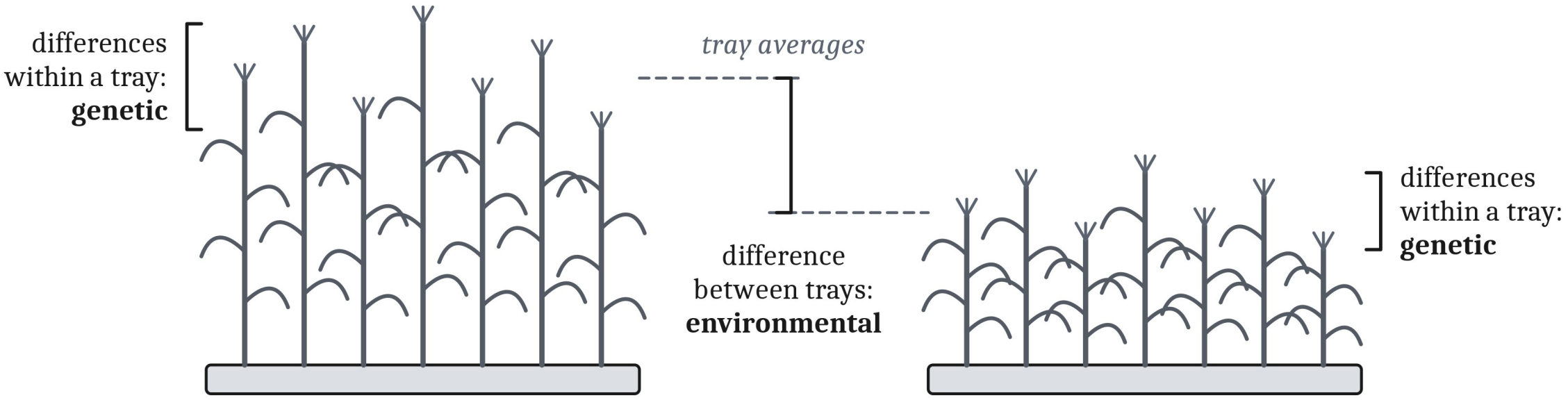
A **genome-wide association study** tests millions of common genetic variants, in very many people, for association with a trait. A **polygenic score** is the sum of a person's genetic variants, each weighted by its association with a trait in a genome-wide association study.

Source of the estimate	Share of the differences in scores
Twin studies	about one half
Common variants measured, in 269,867 people of European ancestry	about a fifth
A polygenic score built from those variants	up to about 5%

Within Groups and Between Them

Content note: claims about racial differences in intelligence. How heritable a trait is within groups gives no evidence about what causes a difference between groups, whatever the size of the heritability. Neither the logic nor the direct evidence supports the claim that the Black–white difference in average scores is genetic.

Lewontin's second experiment with corn



after Lewontin (1970), pp. 7–8; plant heights illustrative

Genes, Groups, and Beliefs

Claims about inherited differences have repeatedly been used to make existing social hierarchies look natural.

Researchers in behavior genetics and evolutionary psychology, and their critics, both reject the inference from what is inherited or evolved to what is good (the *naturalistic fallacy*) and the inference from inherited to inevitable, and both accept that a finding's unwelcome uses do not make it false.

Thinking Ahead

The chimpanzee's stones could come from adapted actions, from learned value, or from planning. Only a test that the simpler procedures would fail can tell them apart, and for him, no such test was run.

Level	What we can now say
Computational	Possible futures multiply, and their outcomes are uncertain and delayed. Scores on many kinds of test are correlated, and testing was used to exclude and sterilize people.
Algorithmic	People plan by searching selectively, and depart from the standards of expected value and consistent choice, in ways that can be read as errors or as rules suited to their environments; how long they wait depends on what they expect. Whether one ability or several produce the correlated scores is still open.
Implementational	The prefrontal cortex may bias processing elsewhere; frontal patients can score well and still decide badly; orbitofrontal neurons fire with value. Heritability within a group is not evidence about the causes of differences between groups.

To Discuss: “Intelligence Is 50% Genetic”

In your group, five minutes.

A headline says, “Intelligence is 50% genetic”.

What does that sentence mean, and what does it not mean? What would change the number?

Image Sources

Slide	Image	File	License	Source
3	Three regions in and beside the prefrontal cortex	fig-prefrontal-regions.png	CC BY 4.0	Jon Willits — from the Module 12 reading · own work
4	Items like those in the original Stroop experiments	fig-stroop.png	CC BY 4.0	Jon Willits — from the Module 12 reading, after Stroop (1935) · own work
7	The multiple-demand network	fig-md-network.png	CC BY 4.0	Jon Willits — from the Module 12 reading, after Duncan (2010) and Fedorenko et al. (2013) · own work
11	Lewontin's second experiment with corn	fig-lewontin-seeds.png	CC BY 4.0	Jon Willits — from the Module 12 reading, after Lewontin (1970) · own work